Dust Growth on Superparticles in AthenaK

Validation Document

D3 tests (a)–(e), cost, and R1: the stratified SI with growth

worktree `athenak-worktrees/dust-growth`, branch `dust-growth` (parent `9aaa5639`)

commits `a821cb4d` (growth step, diagnostics, tests) and `e231fc7f` (`nsh_init`, R1 set-up)

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Abstract

Five tests of the per-superparticle growth step described in `dust_growth_implementation.pdf`, all passing: (a) with growth off the code is byte-identical to its parent; (b) uniform growth follows the analytic $S(t)$ with first-order convergence in Δt; (c) fragmentation holds the turbulent cap $S_{\rm frag} = v_{\rm frag}^2/(3\alpha_t c_s^2)$ to 0.2%; (d) the drift term reproduces the Nakagawa–Sekiya–Hayashi relative velocities of three species to 10^{-12} in 2D, in 3D with orbital advection and in 3D without; (e) the stopping times survive a restart bitwise and pass through AMR particle splitting unchanged. The growth step costs 1–8% per cycle. R1, the stratified streaming instability in a thin 3D box at $1280/H$ with $Z = 0.02$, $\Pi = 0.05$ and grains starting at $S = 10^{-3}$ (below the clumping threshold), shows the effect growth has: the grains reach the fragmentation cap $S \approx 0.33$ within $\sim 55\,\Omega^{-1}$, and the particle density exceeds the Hill density from $t = 82\,\Omega^{-1}$ to the end of the run ($300\,\Omega^{-1}$, peak $1920\rho_{g0}$), while the identical run without growth never exceeds $2.9\rho_{g0}$.

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1 Summary

test	result	status
(a) growth off	all data outputs byte-identical to the parent (2D NSH, 3D AMR on 4 ranks, YJ14 stratified on 4 ranks)	**PASS**
(b) uniform growth	error 2.8×10^{-3} at $C = 0.45$, halving with Δt (order 0.998, 0.999)	**PASS**
(c) fragmentation cap	S held within -0.22% / $+0.12\%$ of $S_{\rm frag}$; reached at $t = 8.76$ (law: 8.74)	**PASS**
(d) drift vs NSH	three species, three geometries: relative error $\leq 1.2 \times 10^{-12}$	**PASS**
(e) restart, AMR	restart bitwise; AMR run with splitting completes on 4 ranks; split keeps $\max S$, $\min S$ bitwise	**PASS**

1.1 Reproducing

From the repository root of the worktree, with the new binary in `build/src/athena` and a baseline binary of the parent commit (default: the `si-strat-amr` worktree's build, which is at `9aaa5639`):

```
scripts/analysis/dust_growth_tests.sh all        # runs -> validation/run/dust_growth/
python3 scripts/analysis/dust_growth_tests.py    # checks; PASS/FAIL per test
julia scripts/analysis/dust_growth_tests_fig.jl  # validation/figures/dust_growth_tests.png
```

The suite takes about 5 minutes on 4 Mac cores. AthenaK's command line can only replace parameters that already exist in an input file (new parameters or blocks are a fatal error), so the shell script writes a patched copy of the input into every run directory (`patch_input`).

2 Test (a): growth off is the parent code

Purpose. The growth task is added to every dust run; with `growth = none` it must change nothing.

Set-up. The parent binary and the new binary run three existing inputs with identical patched copies:

- `dust_nsh.athinput`: 2D r–z shearing box, three species, NSH equilibrium, to $t = 2$ (1 rank);
- `dust_nsh_shear3d_amr.athinput`: 3D shear-periodic NSH with an AMR refine and derefine event and particle splitting, one orbit (4 ranks);
- `streaming_strat_yj14.athinput`: 3D stratified SI, WENO-Z + Roe, 60 cycles with every output (hst, phst, dust density and gas VTK, float64 particle VTK, restart) written frequently (4 ranks).

Result. Every data file is byte-identical: 4/4, 4/4 and 23/23 files. The two restart files of the YJ14 case differ only in their parameter block, where the new binary echoes the defaulted line `growth = none # Default value added at run time` (76 bytes); the binary payload after `<par_end>` is identical. After the `nsh_init` option was added to `streaming_strat`, the YJ14 comparison was repeated (20/20 data files identical), confirming that the generator is unchanged with the option off.

3 Test (b): uniform growth

Set-up. `inputs/dust/dust_growth_box.athinput`: 2D r–z shearing box, 32^2 cells, $L = 1\,H$, periodic, one species on the quiet lattice of the `dust_nsh` generator (1 particle per cell), $\epsilon = \rho_d/\rho_{g0} = 1$, no pressure gradient ($\eta = 0$) and no back-reaction. Gas and dust stay exactly at rest, so the drift term is zero, and the lattice makes the TSC density exactly 1 at every particle. $S_0 = 10^{-3}$, $\alpha_t = 10^{-3}$, $v_{\rm frag} = 10\,c_s$ (no fragmentation), $t_{\rm end} = 20\,\Omega^{-1}$.

Expected. With $\Delta v = \sqrt{3\alpha_t S}c_s$, the law $dS/dt = \sqrt{\pi/8}\,\Omega\,(\rho_d/\rho_{g0})\,\Delta v/c_s$ becomes $d\sqrt{S}/dt = \frac{1}{2}\sqrt{\pi/8}\,(\rho_d/\rho_{g0})\sqrt{3\alpha_t}\,\Omega$, so

$$S(t) = \left[\sqrt{S_0} + \tfrac{1}{2}\sqrt{\pi/8}\,\tfrac{\rho_d}{\rho_{g0}}\sqrt{3\alpha_t}\,\Omega t\right]^2, \qquad S(20) = 0.140518. \tag{1}$$

The forward-Euler split is first order, so the error should halve with Δt.

Result. Three Courant numbers (Fig. 1, first two panels):

C	$S(t=20)$	$\max\lvert S/S_{\rm exact}-1\rvert$	order
0.45	0.140294	2.801×10^{-3}	
0.225	0.140406	1.402×10^{-3}	0.998
0.1125	0.140462	7.014×10^{-4}	0.999

All particles keep identical S (spread 0). The error peaks early, when $d\ln S/dt$ is largest, and the per-cycle cap (5%) never binds. In production the per-step change is $\sim \Delta t/t_{\rm grow} \sim 10^{-5}$, so the splitting error is negligible there.

4 Test (c): the fragmentation cap

Set-up. As (b) with $v_{\rm frag} = 0.01c_s$, so the turbulent cap is $S_{\rm frag} = v_{\rm frag}^2/(3\alpha_t c_s^2) = 0.0333$.

Result. S follows the growth law until $t = 8.76$ (the law reaches $0.99S_{\rm frag}$ at 8.74), then dithers at the cap: for $t > 12$ the mass-weighted mean lies in $[0.033260, 0.033375]$, i.e. -0.22% to $+0.12\%$ of $S_{\rm frag}$, and the overall maximum is 0.033384. The dither amplitude is one step's change at the cap, as designed (Fig. 1, third panel).

5 Test (d): the drift term against NSH

Set-up. `dust_nsh.athinput` (2D r–z, 1 rank) and `dust_nsh_shear3d.athinput` (3D shear-periodic, 4 ranks, with and without orbital advection), growth on with $\alpha_t = 0$ (drift only) and $v_{\rm frag} = 10c_s$, one cycle, float64 particle VTK at cycles 0 and 1. Three species with $\tau_s = 0.01, 0.1, 1$ and $\epsilon_s = 0.2, 0.5, 0.3$, $\eta v_K = 0.05c_s$, $q = 3/2$, lattice placement at the exact multi-species NSH equilibrium.

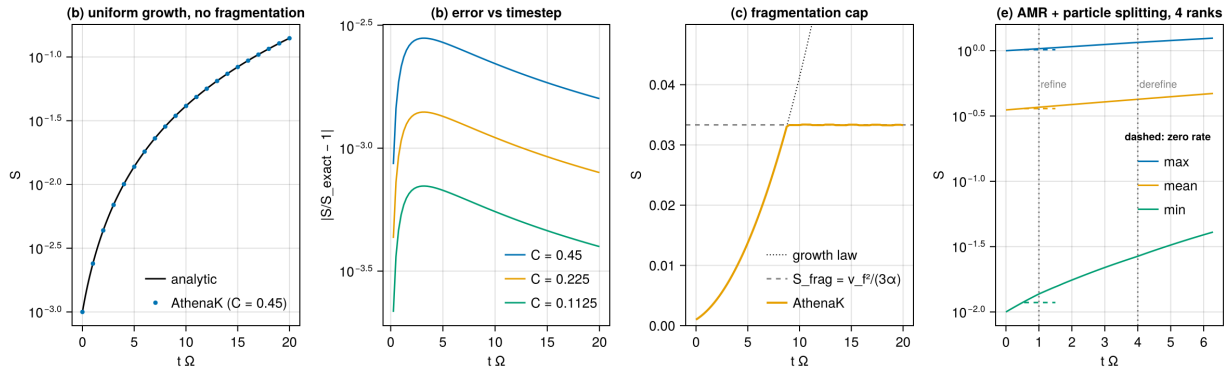


Figure 1: (b) Mass-weighted mean Stokes number against the analytic law, and its error at three Courant numbers. (c) With $v_{\rm frag} = 0.01 c_s$ the growth stops at $S_{\rm frag} = 0.0333$. (e) The 3D AMR NSH run with growth on 4 ranks (solid: max, mass-weighted mean, min of S; dotted lines: the refine and derefine events); dashed: the restart at $t \approx 0.55$ with a zero growth rate, flat through the split at $t \approx 1.05$.

Expected. Over the first cycle each particle's $t_{\rm s}$ changes by $\Delta t \sqrt{\pi/8}\,(\rho_d/\rho_{g0})\,\frac{1}{2}|\mathbf{v}_s - \mathbf{u}|/c_s$ with $\rho_d = \sum_s \epsilon_s \rho_{g0} = 1$ and $|\mathbf{v}_s - \mathbf{u}|$ from the multi-species NSH linear system, solved independently in the checker (`dust_growth_tests.py`, `nsh_dv`).

Result. Measured $(t_{\rm s}^{(1)} - t_{\rm s}^{(0)})/\Delta t$, mean over each species, with the largest relative deviation of any particle:

geometry	species 1	species 2	species 3
2D r–z	$1.6819390297 \times 10^{-4}$	$1.6797615195 \times 10^{-3}$	$1.0029805021 \times 10^{-2}$
	(2.3×10^{-13})	(3.2×10^{-14})	(7.2×10^{-13})
3D, FARGO	same digits	same digits	same digits
	(3.0×10^{-13})	(7.7×10^{-14})	(3.7×10^{-13})
3D, no FARGO	same digits	same digits	same digits
	(5.2×10^{-14})	(8.9×10^{-14})	(1.2×10^{-12})
expected	$1.6819390297 \times 10^{-4}$	$1.6797615195 \times 10^{-3}$	$1.0029805021 \times 10^{-2}$

This checks the gather of the gas velocity, the pairing of velocity components in the 2D r–z box (azimuthal = third component), and, in the no-FARGO case, the background-shear correction applied at the stencil cells, which is exact because TSC weights reproduce linear functions.

6 Test (e): restart and AMR

6.1 Restart

Set-up. The test box with drift ($\eta v_K = 0.05 c_s$, back-reaction on, $\alpha_t = 10^{-3}$), run to $t = 4$ writing a restart file at $t \approx 2$; a second run restarts from that file and continues to $t = 4$. (A first version restarted from a run stopped at $t_{\rm lim} = 2$; its shortened last step changes the timestep sequence, so that comparison is not bitwise by construction.)

Result. The final float64 particle VTK files of the straight and restarted runs are byte-identical.

6.2 AMR and particle splitting

Set-up. `dust_nsh_shear3d_amr.athinput` with growth ($\alpha_t = 10^{-3}$, drift on): the interior bar refines at $t = 1$ (every particle of a refining block becomes 8 children) and derefines at $t = 4$; 4 ranks; restart file at $t \approx 0.55$. *Split check*: restart from that file with $\alpha_t = 0$ and the drift term off, so the growth rate is exactly zero, and run through the refine event to $t = 1.5$. By $t = 0.55$ every t_s has grown away from its species value, so a split that reset `IPTS` to the species table, or lost it, would show.

Result. The AMR run completes (98 304 → 270 336 particles, $\langle S \rangle$ 0.352 → 0.471). In the split check $\max S$ and $\min S$ stay bitwise constant through the split at $t = 1.05$ ($\min S = 0.0118038$ against the species value 0.01); the mass-weighted columns move by at most 7.7×10^{-13}, which is the round-off of summing 2.75 times as many particles of one eighth the mass.

Not a test: AMR against a uniform mesh. Comparing this AMR run with a uniform-mesh twin was the first plan, but the two diverge right after the refine event whether or not growth is on: refining the NSH lattice makes the particle-mesh deposit non-uniform (the maximum dust density jumps from 1.000 to 1.09–1.33 while refined, and to 1.031 after derefinement), and the NSH errors of the unmodified parent at the end of this input are 10^{-7}–10^{-5}, not the round-off its header claims. This is a pre-existing property of the AMR lattice test, unrelated to growth, and has been flagged separately. The same input with `<problem>/amr_test = false` segfaults, because the generator then enrols no refinement function while the input still asks for `method = user`; the uniform twin was instead run with the refinement window moved past the end of the run.

7 Cost

The growth step adds one particle-mesh deposit, its additive and copy exchanges, and one pass over the particles per cycle.

case	machine	without growth	with growth
YJ14 box ($32^2 \times 16$, 16k particles), 300 cycles	Mac, 4 ranks	1.453 s	1.466 s (+0.9%)
R1 box ($256 \times 4 \times 256$, 262k particles)	h-2, 64 ranks	25.2 s	27.1 s (+7.5%)
		(per 1000 cycles)	

The R1 comparison is between the `nogrowth` run (`species_fixed`) and the `fid` run (`dynamic`), so it also includes the two per-cycle stopping-time reductions of `dynamic` mode.

8 R1: the stratified SI with growth

8.1 Set-up (approved October 19)

`inputs/dust/streaming_strat_growth_r1.athinput`, in the style of Lim et al. (2025, ApJ 981, 160) but quasi-2D in a thin 3D box, because the dust module and `streaming_strat` are 3D-only and the 2D r–z shearing box has no vertical gravity:

box	$0.2H \times 4$ cells $\times\, 0.2H$ at $1280/H$ ($256 \times 4 \times 256$), 64 MeshBlocks of $32 \times 4 \times 32$; uniform mesh
boundaries	shear-periodic x with orbital advection (FARGO), periodic y and z
gas	isothermal, WENO-Z + Roe, $C = 0.3$, imex2+ drag
dust	$\Pi = 0.05$, $Z = 0.02$, one species, $S_0 = 10^{-3}$, Gaussian layer $H_{p0} = 0.025H$ with NSH velocities for the local $\epsilon(z)$ (midplane $\epsilon = 0.80$), 1 particle per cell (262 144)
growth	$\alpha_t = 10^{-4}$, $v_{\rm frag} = 0.01 c_s$ ($S_{\rm frag} = 0.33$), drift on
runs	`fid` (growth + back-reaction) and `nogrowth` ($S = 10^{-3}$ fixed); a no-back-reaction control is available (`CASE=nobr`) but was not requested
machine	h-2, jobs 276 (`fid`) and 277 (`nogrowth`), 64 ranks each, run directories `/data/limjay/runs/si_growth_r1/{fid,nogrowth}`, source `/data/limjay/athenak_growth`
length	$t = 300\,\Omega^{-1}$ ($\approx$48 orbits), $\Delta t = 2.2 \times 10^{-4}$ (1.35×10^6 cycles); restart files every $100\,\Omega^{-1}$
outputs	hst and phst every $1\,\Omega^{-1}$, dust density every 2, gas every 20, particle VTK with `ptaus` every 10

Without growth, $S = 10^{-3}$ at $Z = 0.02$ lies below the strong-clumping threshold (Li & Youdin 2021: $Z_{\rm crit}(10^{-3}) \approx 0.035$; Lim et al. 2025 found no strong clumping at $\tau_s = 0.001$, $Z = 0.02$ even at $5120/H$). The question R1 answers is whether growth carries the system across the threshold.

8.2 Smoke test at 320/H

The same input at $320/H$ ($64 \times 4 \times 64$, 4 Mac ranks) to $t = 60\,\Omega^{-1}$ (Fig. 2): stable throughout; the mass-weighted mean S goes $10^{-3} \to 0.017$ ($t = 25$) $\to 0.17$ ($t = 60$); the largest particles reach the cap 0.33 by $t \approx 55$; the maximum dust density climbs from 1.5 to 21 as the layer settles and begins to clump. Particles high in the layer, where ρ_d is small, grow slowly (minimum S only 0.002 at $t = 60$), so the size distribution is broad. The cap of 5% per cycle never bound; fragmentation acted on 7.8×10^6 particle-steps once the largest grains reached the cap.

8.3 Results

Both runs reached $t = 300\,\Omega^{-1}$ (exit 0): `nogrowth` in 9.4 h and `fid` in 19.7 h of wall time. `fid` started at 27 s per 1000 cycles, like `nogrowth` (25 s), but slowed to 42 s once clumps formed, presumably because the dense regions load a few ranks heavily (one MeshBlock per rank, uniform mesh). Figure 3 shows the growth run at $t = 300$ and both runs' maximum density.

Growth. The midplane grains grow first, where ρ_d is largest. The mass-weighted mean Stokes number passes 0.01 at $t = 19$, 0.03 at $t = 35$ and 0.1 at $t = 52\,\Omega^{-1}$; the largest grains reach the turbulent fragmentation cap $S_{\rm frag} = 0.33$ at $t = 53$, and half of the mass has $S > 0.1$ at $t = 55$ (90% at $t = 124$). From then on the size distribution is a narrow spike at the cap (panel c): at $t = 300$ the mean is 0.296 and 99% of the mass has $S > 0.1$. Grains high in the layer, where ρ_d is small, lag far behind (the smallest is still at $S = 0.011$ at $t = 300$).

Clumping. As the grains grow they settle (particle scale height $0.025H$ at $t = 0$, $0.018H$ at 50, $0.011H$ at 100, $0.008H$ at 300) and the SI develops: the maximum particle density passes $10\rho_{g0}$ at

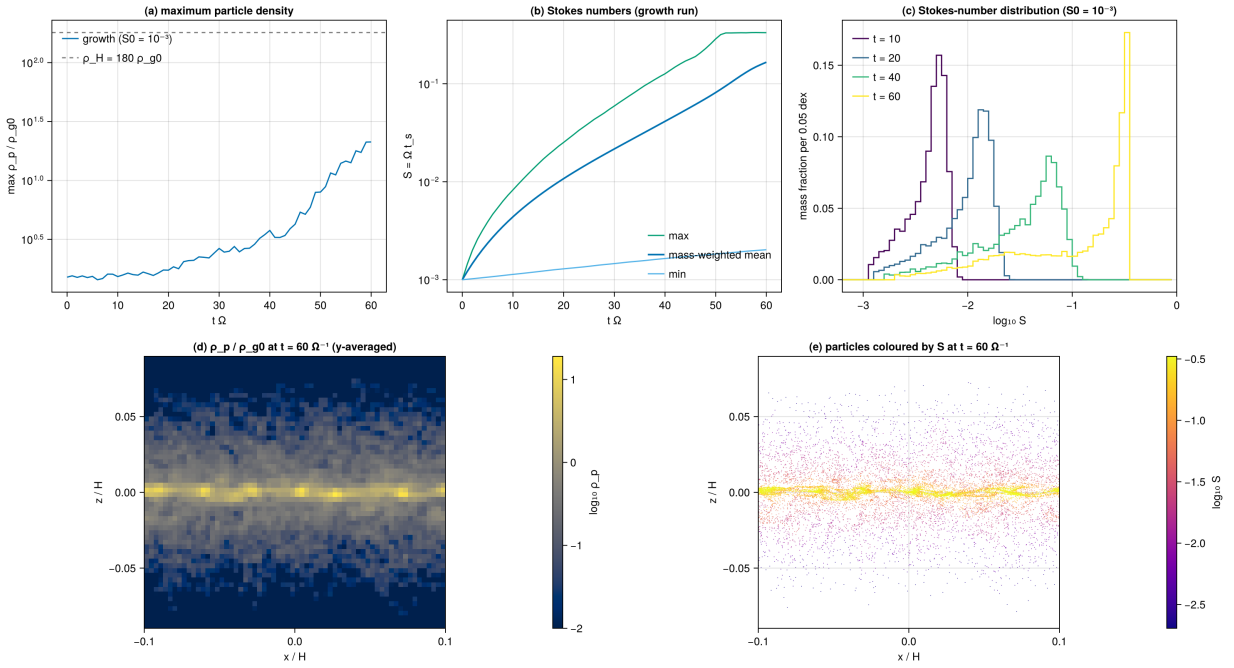


Figure 2: R1 set-up at $320/H$ (smoke test, growth run only). (a) Maximum dust density with the Hill density $180\rho_{g0}$ of Lim et al. (2025); (b) Stokes-number statistics; (c) mass distribution of S; (d) y-averaged dust density and (e) particles coloured by S at $t = 60\,\Omega^{-1}$.

$t = 46$, $100\rho_{g0}$ at $t = 72$ and the Hill density $\rho_H = 180\rho_{g0}$ (Lim et al. 2025) at $t = 82\,\Omega^{-1}$, 13 orbits after the start. It stays above ρ_H for the rest of the run (217 of the 219 outputs after $t = 82$; the lowest is $175\rho_{g0}$), first exceeds $1000\rho_{g0}$ at $t = 194$, and peaks at $1920\rho_{g0}$ at $t = 199$; the median over $t \geq 150$ is $605\rho_{g0}$. At $t = 300$ the layer is a set of tangled, strongly stirred filaments with one dense clump near $x = +0.05H$ (panels d, e). In this quasi-2D box the dense structures are axisymmetric rings, as in the 2D runs of Lim et al. (2025).

The control. With $S = 10^{-3}$ fixed, the maximum particle density never exceeds $2.9\rho_{g0}$ in $300\,\Omega^{-1}$, and the layer is still settling slowly ($H_p = 0.025H \to 0.015H$; the settling time of $S = 10^{-3}$ grains is $\sim 10^3\,\Omega^{-1}$). This agrees with Lim et al. (2025), who found no strong clumping at $\tau_s = 0.001$, $Z = 0.02$ even at $5120/H$ over $10^4\,\Omega^{-1}$.

Growth-step diagnostics. `DUST_GROWTH_SUMMARY` of `fid`: of the 3.6×10^{11} particle-steps, 40% were fragmenting (the particles dithering at the cap) and 2.6×10^5 (7×10^{-7}) hit the 5% per-cycle cap `growth_maxfrac`. This is the first run in which the cap bound at all. The likely cause is small grains ($S \sim 0.005$) swept into clumps of $\rho_p \sim 10^3\rho_{g0}$, where the growth rate exceeds 5% per step: for those particle-steps the growth was set by the cap (and so by Δt) rather than by the law. The minimum Stokes number dips briefly at $t \approx 205$, 250 and 265 (panel b), consistent with a few small grains fragmenting in the strongly perturbed gas around the clumps and regrowing. Neither affects the bulk of the mass, which sits at the cap.

Data. Run directories `/data/limjay/runs/si_growth_r1/{fid,nogrowth}` on h-2 (0.89 and 0.86 GB); archived on October 20 as `<run>_meta.tar` (histories, logs, input, binary), `<run>_data.tar`

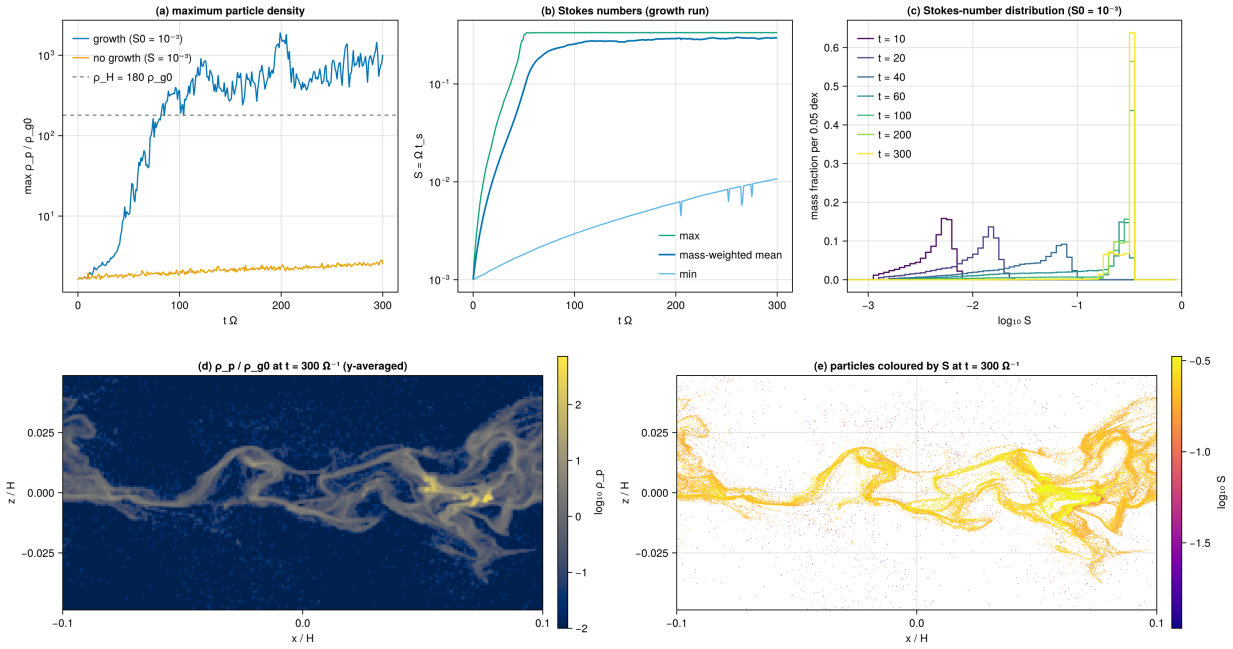


Figure 3: R1 at $1280/H$. (a) Maximum particle density of the growth run (blue) and the no-growth control (orange), with the Hill density $180\rho_{g0}$; (b) Stokes-number statistics of the growth run; (c) mass distribution of S at several times; (d) y-averaged particle density and (e) particles coloured by S at $t = 300\,\Omega^{-1}$.

(restart, mesh and particle VTK files) and `<run>.sha256` in `/data/limjay/archive_dust_growth` for transfer to Lou (checksums verified; the tar contents add up byte for byte to the run directories). The figure is made by `julia scripts/analysis/si_growth_r1_fig.jl <fid dir> <nogrowth dir> <out.png>` from the `phst` files, the particle VTK files at $t = 10, 20, 40, 60, 100, 200, 300$ and the dust density at $t = 300$.

9 Notes from setting up

- A new worktree starts with an empty `kokkos` submodule. The configured submodule URL is a local path, so it initialises without network access with `git -c protocol.file.allow=always submodule update -init kokkos`.
- h-2 has no `rsync`; the source was copied with `tar` over `ssh`. A first copy with `-exclude './build*'` also dropped `kokkos/cmake/build_env_info.cmake` and `src/mesh/build_tree.cpp` (bsdtar matches the pattern below the top level); the missing files were copied and the source list checked against `git ls-files`.